US012401941B2

# (12) United States Patent
Ho

(10) Patent No.: US 12,401,941 B2
(45) Date of Patent: Aug. 26, 2025

(54) **SOUND GENERATING MODULE**

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( * ) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 279 days.

(21) Appl. No.: **18/181,586**

(22) Filed: **Mar. 10, 2023**

(65) **Prior Publication Data**

US 2024/0267672 A1 Aug. 8, 2024

(30) **Foreign Application Priority Data**

Feb. 7, 2023 (TW) ................................ 112104325

(51) **Int. Cl.**
***H04R 1/28*** (2006.01)
***H04R 1/02*** (2006.01)

(52) **U.S. Cl.**
CPC ........... ***H04R 1/2811*** (2013.01); ***H04R 1/021*** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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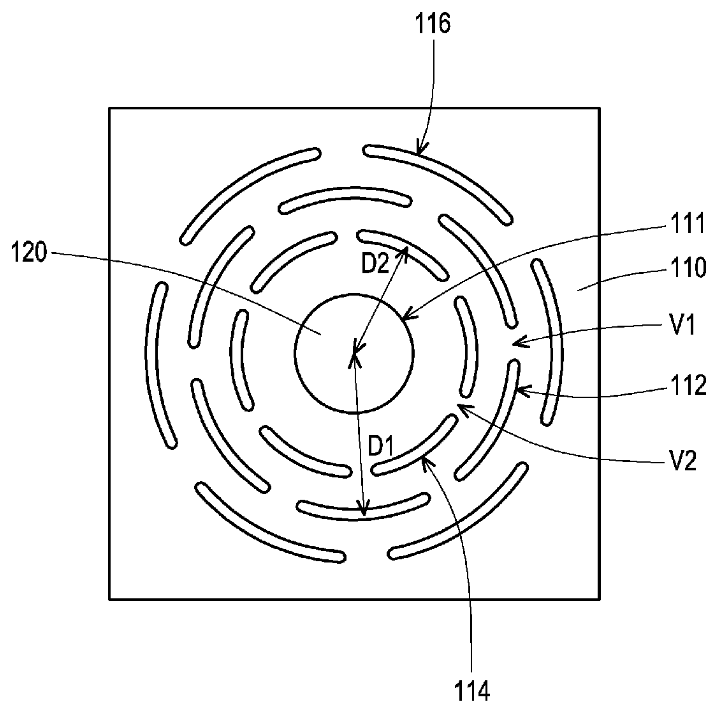
*Primary Examiner* — Paul W Huber

(57) **ABSTRACT**

The invention provides a sound generating module including a baffle and a speaker. The baffle has a containing opening. The speaker is disposed on the baffle and in the containing opening. A sound spectrum of the speaker itself has at least one peak. A wavelength corresponding to the at least one peak includes λ. The baffle has at least one first sound hole. A distance from a center of the first sound hole to a center of the speaker is greater than or equal to λ/4 and is less than or equal to λ/2.

**19 Claims, 6 Drawing Sheets**




100a

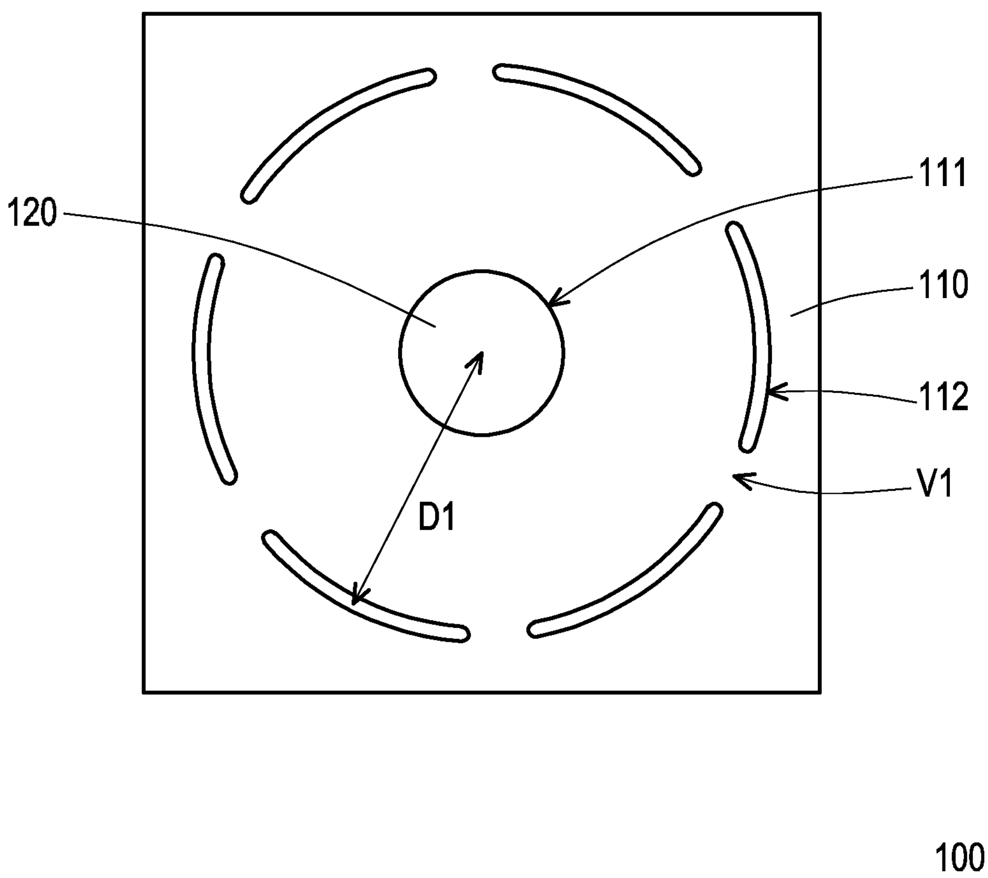



FIG. 1

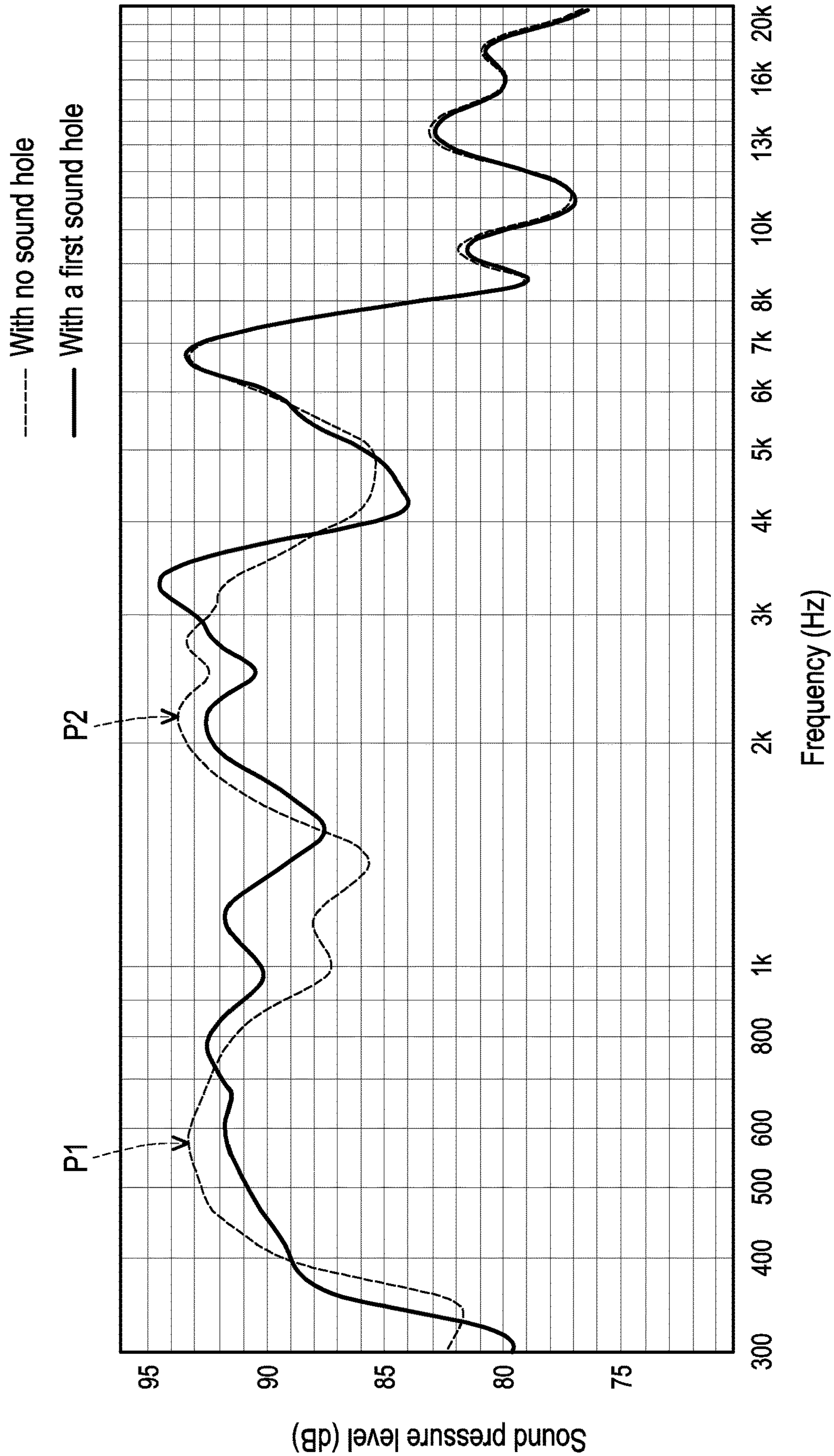

With no sound hole
With a first sound hole
P1
P2
95
90
85
80
75
Sound pressure level (dB)
300
400
500
600
800
1k
2k
3k
4k
5k
6k
7k
8k
10k
13k
16k
20k
Frequency (Hz)

FIG. 2

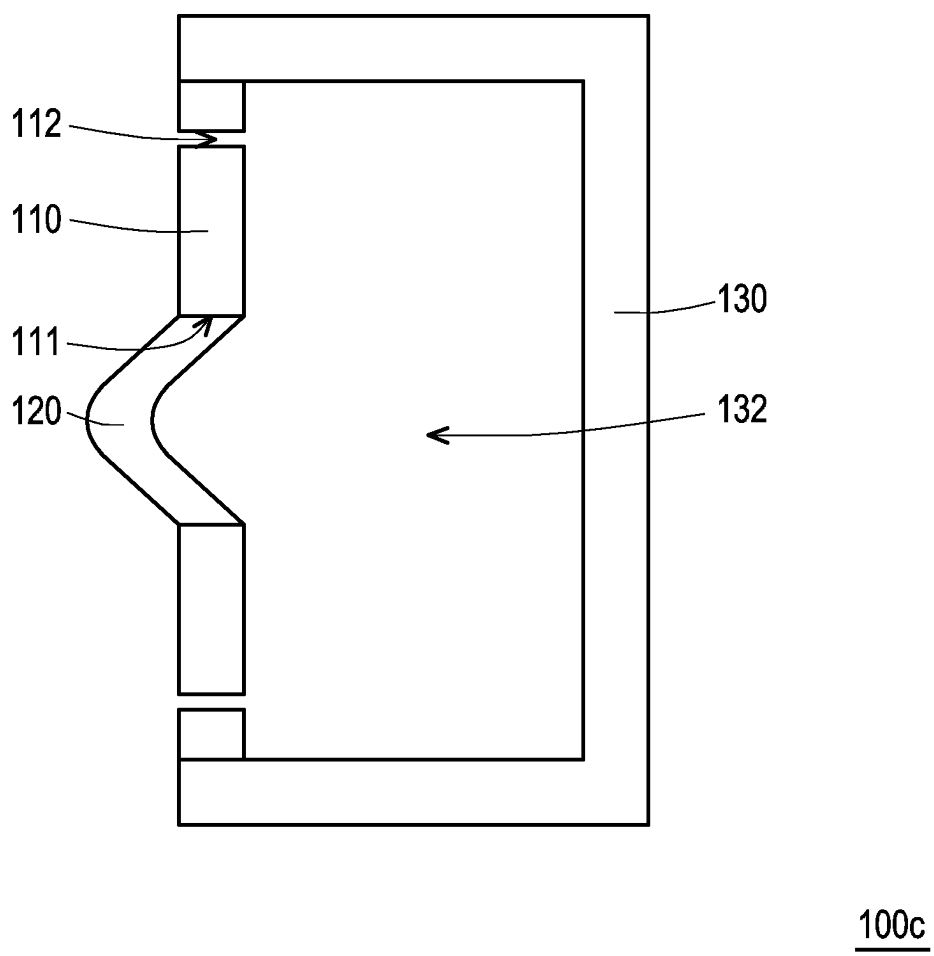



FIG. 3

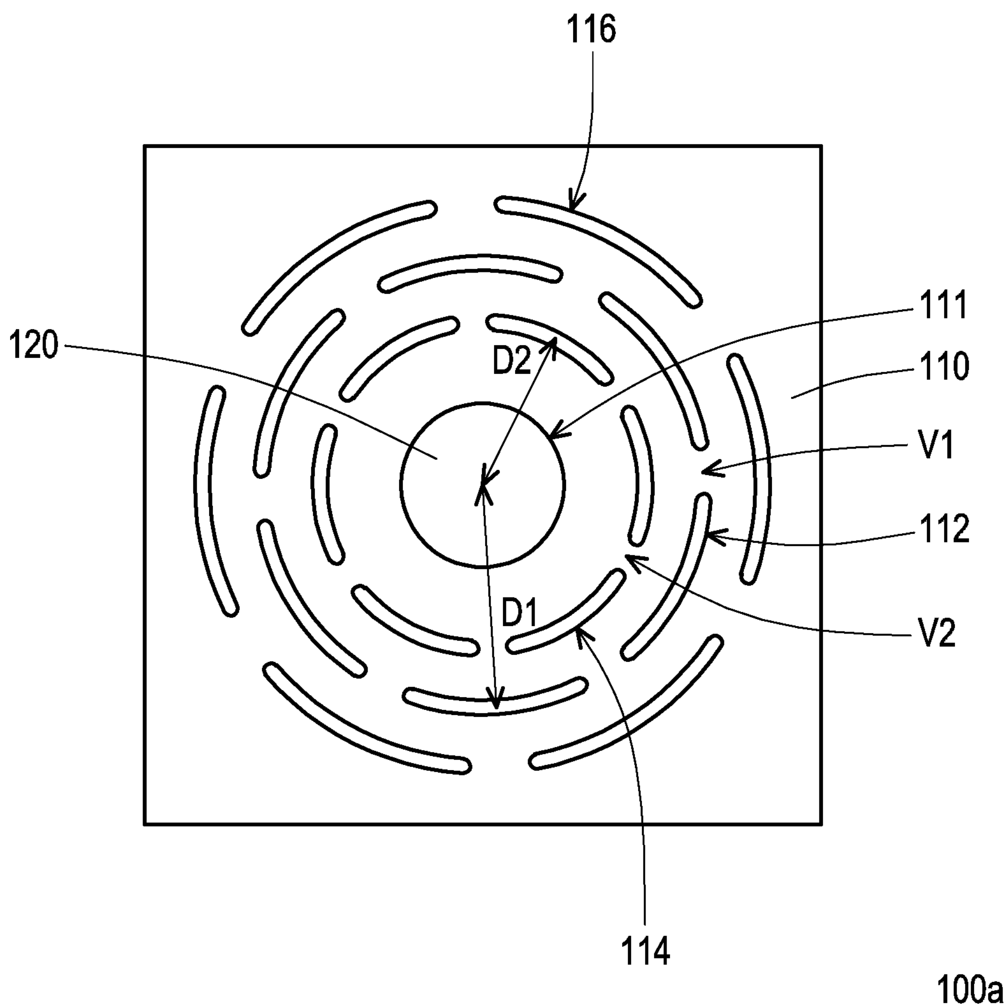



FIG. 4

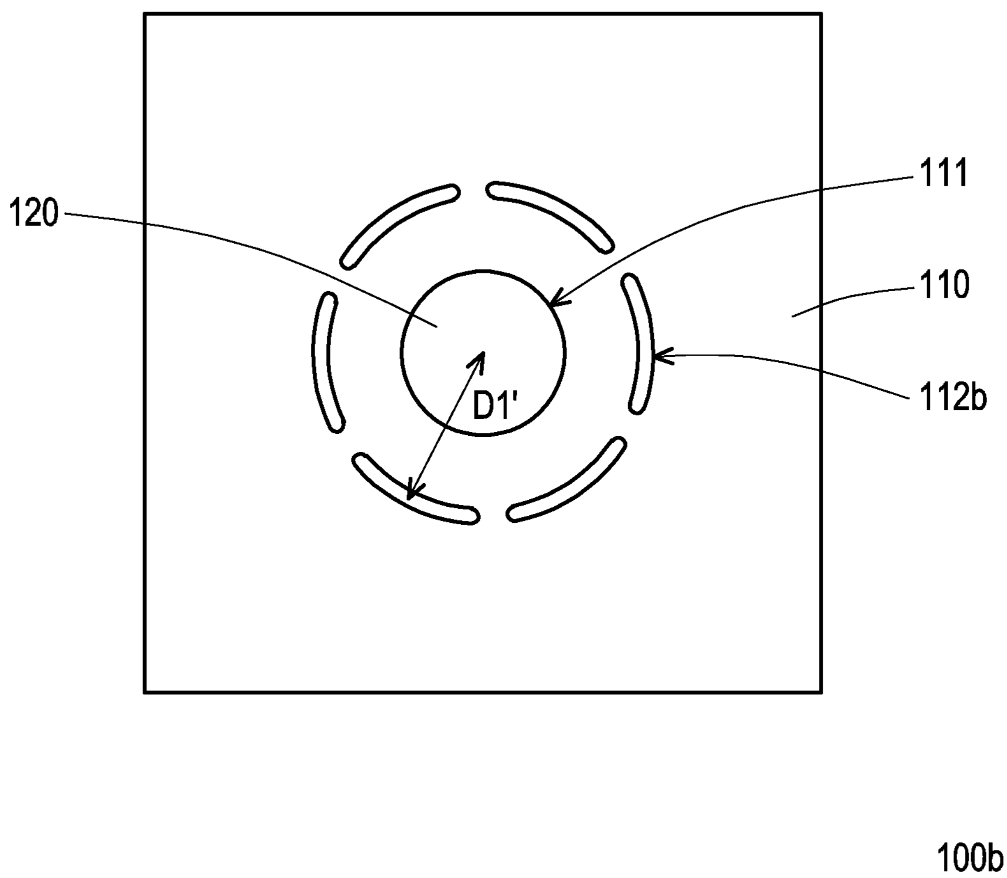



FIG. 5

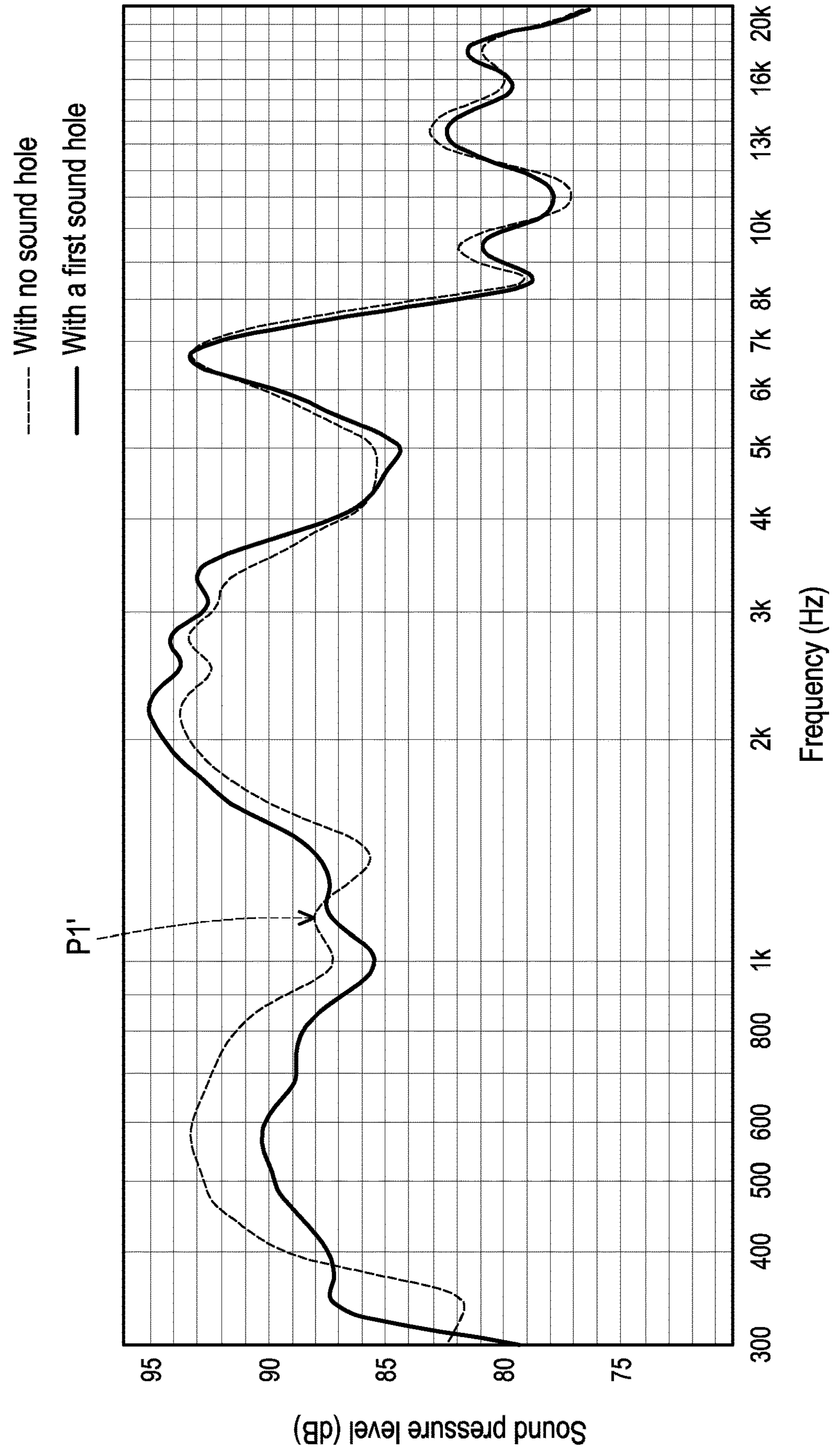

With no sound hole
With a first sound hole
P1'
95
90
85
80
75
Sound pressure level (dB)
300
400
500
600
800
1k
2k
3k
4k
5k
6k
7k
8k
10k
13k
16k
20k
Frequency (Hz)

FIG. 6

# SOUND GENERATING MODULE

## CROSS-REFERENCE TO RELATED APPLICATION

This application claims the priority benefit of Taiwan application serial no. 112104325, filed on Feb. 7, 2023. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

## BACKGROUND

### Technical Field

The disclosure relates to a sound generating module.

### Description of Related Art

A sound generating module may include a speaker, a baffle, or further a sound box, which may emit a corresponding sound according to an electrical signal to restore a sound recorded by a microphone. Therefore, the sound generating module has become one of the important electronic products or electronic components in the current multimedia era.

For speaker systems on the market, whether being a full sound-box-type speaker or a baffle-type speaker, it is difficult for frequency response curves of the speakers to be flat. In order to achieve making the response curve flat, adjusting an electronic equalizer (EQ) or crossover is mostly adopted in the related art. However, such a processing method is likely to cause disadvantages such as early sound distortion, reduction of efficiency and sensitivity, and the like.

## SUMMARY

The disclosure provides a sound generating module, which can achieve the effect of adjusting the frequency response curve in a way of acoustic filtering, and can effectively avoid sound distortion.

An embodiment of the disclosure provides a sound generating module, including a baffle and a speaker. The baffle has a containing opening, and the speaker is disposed on the baffle and in the containing opening. A sound spectrum of the speaker itself has at least one peak, and a wavelength corresponding to the at least one peak includes λ. The baffle has at least one first sound hole, and a distance from a center of the first sound hole to a center of the speaker is greater than or equal to λ/4 and is less than or equal to λ/2.

In the sound generating module of the embodiment of the disclosure, the first sound hole is designed on the baffle, and the distance from the center of the first sound hole to the center of the speaker is greater than or equal to λ/4 and is less than or equal to λ/2. Therefore, the sound waves of a wavelength near λ and generated at the front side and the back side of the speaker may generate a destructive interference at the first sound hole to flatten the frequency response curve. Therefore, the sound generating module of the embodiment of the disclosure can achieve the effect of adjusting the frequency response curve in a way of acoustic filtering, and can effectively avoid sound distortion.

## BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a schematic front view of a sound generating module according to an embodiment of the disclosure.

FIG. **2** is a view of frequency response curves of a sound generating module control group with no sound hole in a baffle and the sound generating module with a first sound hole in FIG. **1**.

FIG. **3** is a schematic cross-sectional view of a sound generating module according to another embodiment of the disclosure.

FIG. **4** is a schematic front view of a sound generating module according to another embodiment of the disclosure.

FIG. **5** is a schematic front view of a sound generating module according to another embodiment of the disclosure.

FIG. **6** is a view of frequency response curves of a sound generating module control group with no sound hole in a baffle and the sound generating module with the first sound hole in FIG. **5**.

## DESCRIPTION OF THE EMBODIMENTS

FIG. **1** is a schematic front view of a sound generating module according to an embodiment of the disclosure, and FIG. **2** is a view of frequency response curves of a sound generating module control group with no sound hole in a baffle and a sound generating module with a first sound hole in FIG. **1**. Please refer to FIG. **1** and FIG. **2**. A sound generating module **100** of this embodiment includes a baffle **110** and a speaker **120**. The baffle **110** has a containing opening **111**, and the speaker **120** is disposed on the baffle **110** and in the containing opening **111**. In this embodiment, the speaker **120** is, for example, a transducer which converts electrical energy into acoustic energy. A sound spectrum of the speaker **120** itself is similar to the frequency response curve of the sound generating module with no sound hole in the baffle in FIG. **2**, in which, the sound spectrum of the speaker **120** itself has at least one peak (for example, a peak **P1** and a peak **P2** in FIG. **2**, a wavelength corresponding to the at least one peak includes λ (for example, the wavelength corresponding to the peak **P1** is λ)). The baffle **110** has at least one first sound hole **112** (a plurality of the first sound holes **112** are taken as an example in FIG. **1**), and a distance D**1** from a center of the first sound hole **112** to a center of the speaker **120** is greater than or equal to λ/4 and is less than or equal to λ/2.

In the sound generating module **100** of this embodiment, the first sound hole **112** is designed on the baffle **110**, and the distance D**1** from the center of the first sound hole **112** to the center of the speaker **120** is greater than or equal to λ/4 and is less than or equal to λ/2. Therefore, the sound waves of the wavelength near λ and generated at the front side and the back side of the speaker **120** may generate a destructive interference at the first sound hole **112** to flatten the frequency response curve. Therefore, the sound generating module **100** of the embodiment of the disclosure can achieve the effect of adjusting the frequency response curve in a way of acoustic filtering, and can effectively avoid sound distortion.

Specifically, disposing the first sound hole **112** at ¼ of a wavelength corresponding to 720 Hz may control to start an attenuation below 720 Hz (so the solid curve below 720 Hz is lower than the dashed curve) and at the same time may increase a gain at ⅛ of the wavelength (approximately 700 Hz to 1400 Hz (at ¼ of the wavelength to ⅛ of the wavelength, the solid curve is higher than the dashed curve)) to achieve the purpose of physical acoustic filter tuning.

In this embodiment, these first sound holes **112** surround the speaker **120**, and spacings V**1** are between the first sound holes **112**. In this embodiment, the first sound holes **112** are arranged in an equal-spacing ring shape. In addition, in this embodiment, the first sound hole **112** is a through hole penetrating through the front side and the back side of the baffle **110**.

In an embodiment, the distance D**1** from the center of the first sound hole **112** to the center of the speaker **120** is equal to $\lambda/4$, so that a forward sound wave on the front side of the speaker **120** and a reverse sound wave on the back side of the speaker **120** generate a destructive interference with respect to sound waves of a wavelength $\lambda$ at the first sound hole **112**.

In this embodiment, the first sound hole **112** is arc-shaped. For example, an arc shape formed by the first sound hole **112** is an arc shape with the center of the speaker **120** as the center. In addition, in this embodiment, no sound box may be disposed on a side of the baffle **110**, that is to say, the sound generating module **100** of this embodiment is a baffle-type speaker. However, in another embodiment, as shown in the schematic cross-sectional view of the sound generating module shown in FIG. **3**, a sound generating module **100***c* further includes a sound box **130**, in which the baffle **110**, the speaker **120**, and the sound box **130** surround a space **132**. In other words, the sound generating module **100***c* is a sound-box-type speaker.

FIG. **4** is a schematic front view of a sound generating module according to another embodiment of the disclosure. Please refer to FIG. **2** and FIG. **4**. A sound generating module **100***a* of this embodiment is similar to the sound generating module **100** in FIG. **1**, and the differences between the two are as follows. In the sound generating module **100***a* of this embodiment, a wavelength corresponding to another peak P**2** of the sound spectrum of the speaker **120** itself is $\delta$, the baffle **110** has at least one second sound hole **114** (a plurality of the second sound holes **114** are taken as an example in FIG. **4**), and a distance D**2** from a center of the second sound hole **114** to the center of the speaker **120** is greater than or equal to $\delta/4$ and is less than or equal to $\delta/2$, in which $\delta$ is not equal to $\lambda$. In this way, the other peak P**2** in the frequency response curve may also be flattened. In an embodiment, the distance D**2** from the center of the second sound hole **114** to the center of the speaker **120** is equal to $\delta/4$, so that the forward sound wave on the front side of the speaker **120** and the reverse sound wave on the back side of the speaker **120** generate a destructive interference with respect to sound waves of a wavelength $\delta$ at the second sound hole **114**.

In this embodiment, the second sound holes **114** surround the speaker **120**, and spacings V**2** are between the second sound holes **114**. In this embodiment, the second sound holes **114** are arranged in an equal-spacing ring shape. In this embodiment, the distance D**2** from the center of the second sound hole **114** to the center of the speaker is not equal to the distance D**1** from the center of the first sound hole **112** to the center of the speaker. In addition, in this embodiment, the first sound holes **112** surround the second sound holes **114**. In this embodiment, the first sound holes **112** surround the second sound holes **114** in a dislocation manner compared with the second sound holes **114**, and the first sound holes **112** and the second sound holes **114** are arranged in a concentric ring shape.

In this embodiment, the second sound hole **114** is a through hole penetrating through the front side and the back side of the baffle **110**. In this embodiment, the second sound hole **114** is arc-shaped. For example, an arc shape formed by the second sound hole **114** is an arc shape with the center of the speaker **120** as the center.

In an embodiment, the sound spectrum of the speaker **120** itself may also have one or more other peaks, the baffle **110** may also have one or more other sound holes (such as a third sound hole **116** in FIG. **4**), and distances from centers of other sound holes to the center of the speaker **120** are greater than or equal to ¼ of wavelengths corresponding to other peaks and are less than or equal to ½ of the wavelengths corresponding to the other peaks. In an embodiment, the distances from the centers of the other sound holes to the center of the speaker **120** are equal to ¼ of the wavelengths corresponding to the other peaks. That is to say, in other embodiments, sound holes with different distances from the center of the speaker **120** may be designed for different peaks.

FIG. **5** is a schematic front view of a sound generating module according to another embodiment of the disclosure, and FIG. **6** is a view of frequency response curves of a sound generating module control group with no sound hole in a baffle and a sound generating module with a first sound hole in FIG. **5**. A sound generating module **100***b* of this embodiment is similar to the sound generating module **100** of FIG. **1**, and the differences between the two are that in the sound generating module **100***b* of this embodiment, a distance D**1'** from the center of a first sound hole **112***b* to the center of the speaker **120** is shorter than the distance D**1** in FIG. **1**, and a peak P**1'** corresponding to the design of the first sound hole **112***b* is also different from the peak P**1** corresponding to the design of the first sound hole **112** in FIG. **1**. In this embodiment, disposing the first sound hole **112***b* at ¼ of a wavelength corresponding to 1.22 KHz may control to start an attenuation below 1.22 KHz (so the solid curve below 1.22 KHz is lower than the dashed curve) and at the same time may increase a gain at ⅛ of the wavelength (approximately 1.22 KHz to 2.44 KHz (at ¼ of the wavelength to ⅛ of the wavelength, the solid curve is higher than the dashed curve)) to achieve the purpose of physical acoustic filter tuning.

In summary, in the sound generating module of the embodiment of the disclosure, the first sound hole is designed on the baffle, and the distance from the center of the first sound hole to the center of the speaker is greater than or equal to $\lambda/4$ and is less than or equal to $\lambda/2$. Therefore, the sound waves of a wavelength near $\lambda$ and generated at the front side and the back side of the speaker may generate a destructive interference at the first sound hole to flatten the frequency response curve. Therefore, the sound generating module of the embodiment of the disclosure can achieve the effect of adjusting the frequency response curve in a way of acoustic filtering, and can effectively avoid sound distortion.

What is claimed is:

**1**. A sound generating module, comprising:

a baffle having a containing opening; and

a speaker, disposed on the baffle and in the containing opening, wherein a sound spectrum of the speaker itself has at least one peak, a wavelength corresponding to the at least one peak comprises $\lambda$, the baffle has at least one first sound hole, and a distance from a center of the first sound hole to a center of the speaker is greater than or equal to $\lambda/4$ and is less than or equal to $\lambda/2$, and wherein the at least one peak comprises a peak of the wavelength corresponding to $\lambda$ and another peak of a wavelength corresponding to $\delta$, the baffle further has at least one second sound hole, and a distance from a center of the second sound hole to the center of the speaker is greater than or equal to $\delta/4$ and is less than or equal to $\delta/2$, wherein $\delta$ is not equal to $\lambda$.

**2**. The sound generating module according to claim **1**, wherein the at least one first sound hole is a plurality of first sound holes, the plurality of first sound holes surround the speaker, and spacings are between the plurality of first sound holes.

**3**. The sound generating module according to claim **2**, wherein the plurality of first sound holes are arranged in an equal-spacing ring shape.

**4**. The sound generating module according to claim **1**, wherein the distance from the center of the first sound hole to the center of the speaker is equal to $\lambda/4$, so that a forward sound wave on a front side of the speaker and a reverse sound wave on a back side of the speaker generate a destructive interference with respect to sound waves of the wavelength $\lambda$ at the first sound hole.

**5**. The sound generating module according to claim **1**, wherein the first sound hole is a through hole passing through a front side and a back side of the baffle.

**6**. The sound generating module according to claim **1**, wherein the first sound hole is arc-shaped.

**7**. The sound generating module according to claim **6**, wherein an arc shape formed by the first sound hole is an arc shape with the center of the speaker as a center.

**8**. The sound generating module according to claim **1**, wherein the at least one second sound hole is a plurality of second sound holes, the plurality of second sound holes surround the speaker, and spacings are between the plurality of second sound holes.

**9**. The sound generating module according to claim **8**, wherein the plurality of second sound holes are arranged in an equal-spacing ring shape.

**10**. The sound generating module according to claim **1**, wherein the distance from the center of the second sound hole to the center of the speaker is not equal to the distance from the center of the first sound hole to the center of the speaker.



**11**. The sound generating module according to claim **1**, wherein the at least one first sound hole is a plurality of first sound holes, the at least one second sound hole is a plurality of second sound holes, and the plurality of first sound holes surround the plurality of second sound holes.

**12**. The sound generating module according to claim **11**, wherein the plurality of first sound holes surround the plurality of second sound holes in a dislocation manner compared with the plurality of second sound holes.

**13**. The sound generating module according to claim **11**, wherein the plurality of first sound holes and the plurality of second sound holes are arranged in a concentric ring shape.

**14**. The sound generating module according to claim **1**, wherein the distance from the center of the second sound hole to the center of the speaker is equal to $\delta/4$, so that a forward sound wave on a front side of the speaker and a reverse sound wave on a back side of the speaker generate a destructive interference with respect to sound waves of the wavelength $\delta$ at the second sound hole.

**15**. The sound generating module according to claim **1**, wherein the second sound hole is a through hole passing through a front side and a back side of the baffle.

**16**. The sound generating module according to claim **1**, wherein the second sound hole is arc-shaped.

**17**. The sound generating module according to claim **16**, wherein an arc shape formed by the second sound hole is an arc shape with the center of the speaker as a center.

**18**. The sound generating module according to claim **1**, further comprising a sound box, wherein the baffle, the speaker, and the sound box surround a space.

**19**. The sound generating module according to claim **1**, wherein no sound box is disposed on a side of the baffle.

* * * * *